

Project Title: **Design of IOT Architecture for Battery Charging Management System**

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Project Fund Requested: **30,000 Rs**

Duration of the Project : **4 Months**



Introduction

- Rapid growth of electric vehicles (EVs) increasing demand for efficient charging infrastructure.
- Need for sustainable solutions integrating renewable energy sources like solar power and energy storage systems (ESS).

•Objective:

- Develop an IoT-based architecture to optimize battery charging management.
- Utilize real-time data monitoring and predictive analytics to enhance efficiency.

•Key Benefits:

- Significant reduction in operational costs, achieving up to 88.5% cost efficiency.
- Enhanced battery management through real-time monitoring of critical parameters (voltage, temperature, SOC).

•Market Demand:

- Increasing EV sales necessitating more charging stations.
- Consumer preference shifting towards renewable energy-powered solutions.

•Future Scope:

- Optimize operations and automate maintenance.
- Expand charging infrastructure to meet growing demand.



Research Gap & Problem

- Lack of uniform standards for EV charging stations.
- Limited Integration of Renewable Energy Sources.
- Absence of Comprehensive IoT-Based Charging Management.
- No Holistic Approach to Grid Stability.
- Minimal Real-Time Monitoring and Control Capabilities.

Motivation

- Need of robust and efficient charging infrastructure.
- Integrating renewable energy sources and Automation.
- To reduce operational costs in charging management.
- Innovation with IOT Architecture in battery management technologies.

Novelty

- The proposed five-layered architecture uniquely combines IoT technology.
 - real-time monitoring in battery charging management.
- The PCC with source integration and automation is to optimize energy efficiency.
- The simulations indicate a potential reduction of up to 88% in operational costs.

1.Design IoT Architecture for Battery Charging Management

1. Remote Monitoring and Control
2. Hybrid Energy Source Integration

2.Enhance Grid Stability

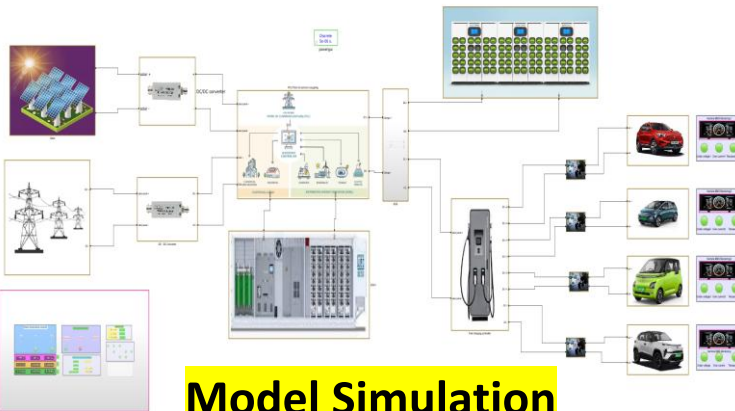
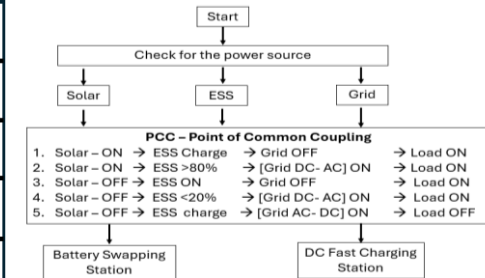
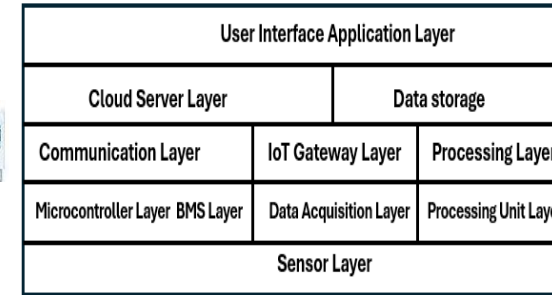
1. Real-Time Data Processing
2. Automated Charging Control
3. Minimize Power Grid Disturbances

3. Achieve High Energy Efficiency

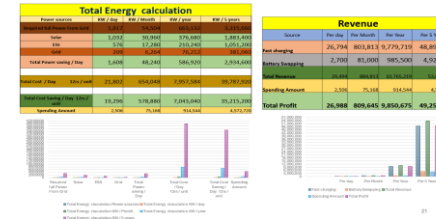
1. Target: 88% Energy Efficiency
2. Optimize Energy Savings
3. Seamless Renewable and Non-Renewable Energy Integration

1. Design Phase

- Integrate Renewable/Non-Renewable Sources
- IOT 5 Layered Architecture
- Create Stable Charging Infrastructure
- Establish Point of Common Coupling (PCC)
- Design Battery Swapping & Fast Charging Stations



Model Simulation



2. Simulation Phase

- MATLAB Simulation
- Energy Efficiency Target: 88.5%
- Node-RED Data Monitoring
- AWS S3 Cloud Storage
- QuickSight Data Visualization

3. Hardware Prototype Phase

- Real-Time Fault Detection
- IoT Architecture Implementation
- Practical Validation of Simulation Results
- Advanced Charging Infrastructure Prototype

Estimated – Energy generation required

Power Generation					
Power Source	Voltage [V]	Current [I]	% of Generation/day	Power [P]- KWh	Power/day
Solar	430	100	60	43	1032
ESS	480	50	30	24	576
Grid	435	20	10	9	209
Total Power Generation per day in KW/24 hr					1817

PCC - Point of common coupling		
PCC will Logically be controlled on availability of source		
Voltage - [V]	430	Output Power/day in KW
Current - [I]	80	
Power - [W] - KW/h	75	
Power Availability	1.17	1772 use / day

ESS - Charging	
No.of batteries	40
Voltage	12
Amper hour - capacity	100
Power / charge - Watt / hr	48000
Power / one cycle charge - KW	48

Fast Charging Station Load								
Cars	Voltage (V)	Power (kW)	Battery Capacity (kWh)	SOC Available	Required Current	Charging Time in hr	Unit charge	Cost
1 - Tata Nexon - EV	400	30.2	30.2	20	75.5	2.47	10	745.83
2- MG ZS EV	400	50	44.5	30	125	1.49	10	743.75
3-Tata Trigo EV	400	16.8	26	50	42	4.40	10	739.58
4- Mahindra eVerito	400	70	60	10	175	1.07	10	747.92
Total Power		167			Total Time	9	Total Revenue	2977

Total Power	1670	KW power can charge / day	Total Time	94 / 96 hr	Total Revenue	26794
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Battery Swapping Station	
Total Batteries	30
Under maintance	0
Avaliable batteries	30
Voltage	12
Current	10
Time / Battery in hr	5
No of Cycle per day	3
Power/Battery - KW	0.12
Total Station KW	18
Total Energy Required for Full Swapping Station per Day - KW	54

Total Batteries charging / day	90
Cost of charge per battery	30
Total Amount Per day	2700

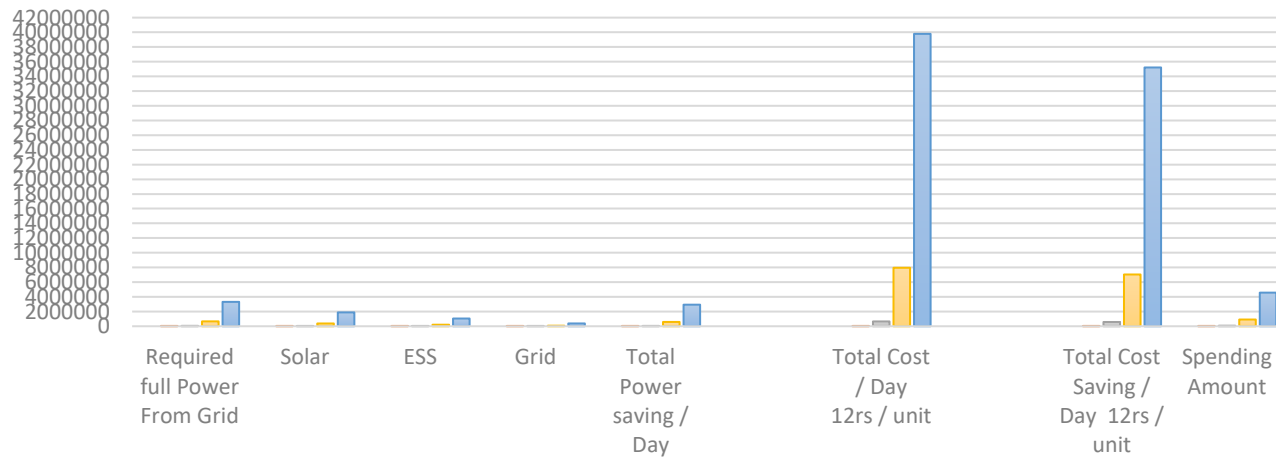
Estimated – Cost and Energy efficiency - Outcomes

Total Energy calculation

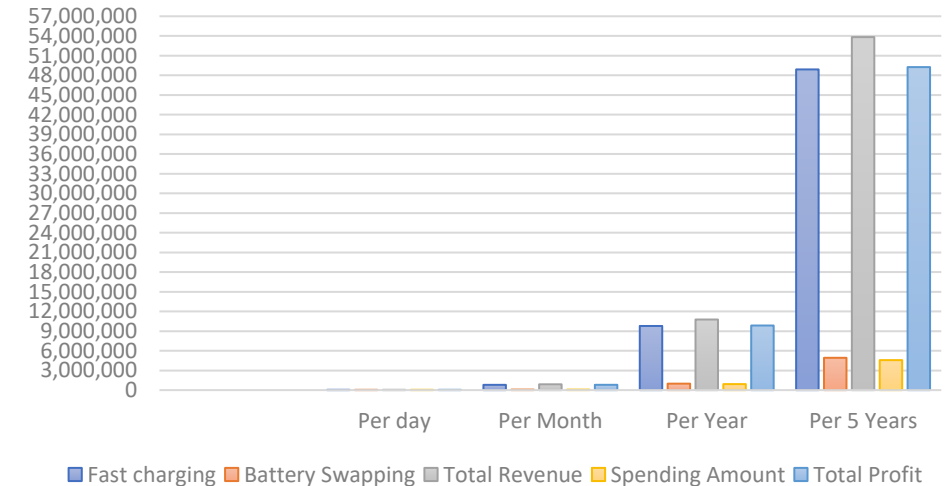
Power sources	KW / day	KW / Month	KW / year	KW / 5 years
Required full Power From Grid	1,817	54,504	663,132	3,315,660
Solar	1,032	30,960	376,680	1,883,400
ESS	576	17,280	210,240	1,051,200
Grid	209	6,264	76,212	381,060
Total Power saving / Day	1,608	48,240	586,920	2,934,600
Total Cost / Day 12rs / unit	21,802	654,048	7,957,584	39,787,920
Total Cost Saving / Day 12rs / unit	19,296	578,880	7,043,040	35,215,200
Spending Amount	2,506	75,168	914,544	4,572,720

Revenue

Source	Per day	Per Month	Per Year	Per 5 Years
Fast charging	26,794	803,813	9,779,719	48,898,594
Battery Swapping	2,700	81,000	985,500	4,927,500
Total Revenue	29,494	884,813	10,765,219	53,826,094
Spending Amount	2,506	75,168	914,544	4,572,720
Total Profit	26,988	809,645	9,850,675	49,253,374



■ Total Energy calculation Power sources
 ■ Total Energy calculation KW / day
 ■ Total Energy calculation KW / Month
 ■ Total Energy calculation KW / year
 ■ Total Energy calculation KW / 5 years



Expected Research Outcome

1

➤ IoT-Enabled EV Charging Prototype

- Functional Battery Charging Management System
- Integration with Cloud Infrastructure
- Real-Time Monitoring Capabilities

2

➤ Research Documentation

- Complete System Architecture Documentation
- Performance Analysis Reports
- Implementation Guidelines

3

➤ Design Patent

- Novel IoT Architecture for Battery Charging
- Integration Method of Renewable/Non-Renewable Sources
- Smart Load Management System

Budget Details Requested – with Justification

Project bill of materials

S.no	Components	Items cost
1	Solar Pannel + Solar charger converter	4500
2	Lead Acid Rechargble Batteries	2600
3	Power Converters	3500
4	Lithium-Ion Batteries + Measurement kit	5400
5	sensors + Switches	5600
6	IOT Microcontrollers	3840
7	Raspberry Pi Kit	4000
8	Tool Kit + Statinonary	1000
Total		30440

Patents with SRM AP Affiliations

1. Patent Filled at Indian Patent office [05.02.2025] - Application no - **202541009491**
2. Hardware Prototype